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From the Editor-in-Chief

I would like to personally apologize for the lack of a graphic cover. That out of the way...

This is the seventy-third volume of *Math Survey*, the official mathematics publication of Stuyvesant High School. I had hoped at the beginning of this year to publish two issues, and because of spectacular enrollment (greatly due to the overwhelming response in Ms. Nelda Latham's math honors class) it looks like it's going to come true. The first issue of *Math Survey* (the one you are currently reading) will be the same quality issue that readers have come to expect from us for 73 years. For the second issue, Jeong, Zardosht (the Associate Editors), and I agreed to publish a special Westinghouse/Intel STS edition, featuring some of our school's finest individual research papers in mathematics, a sample of which is given as the last article of this issue. Look for the full issue later on in the year.

As Editor-in-Chief, I would first like to thank the *Math Survey* staff for their hard work and cooperation throughout the year. Many of our staff members graduated last year, and thus our returning staff members were essential to our publication. In addition to the returning members, many new staff members proved themselves to be equally capable. I would also like to extend my gratitude to all the mathematics teachers in Stuyvesant High School, for I have recently learned to appreciate their guidance. They have all taught the members of our staff, who in turn were able to contribute to publishing this edition of *Math Survey*. Furthermore, I would like to thank Mr. Allen Clancy, our faculty advisor, for his helpful feedback. Finally, I would like to acknowledge the support from all the advertisers, sponsors, and readers of *Math Survey*.

To all who make *Math Survey* possible, I hope that you will enjoy this issue.

Sincerely,



Weikai Lang
Editor-in-Chief

[note: as a sign of the changing times, Math Survey now has its own web site. Please visit us at

www.mathsurvey.cjb.net

Feigenbaum's Constant

Dmitriy Aronov

There is a long-standing tradition in science fiction that the best way to make contact with extraterrestrial civilizations is by means of mathematics. Often, this involves methods such as the transmission of the digits of π , presumably because of its universality. In 1975, however, Mitchell Feigenbaum discovered a constant that is far more universal than π and can be found in many different places.

Many real world systems are modeled using their mathematical equivalents, called the dynamical systems. In these systems, an initial number is constantly changed by some function, and the n th value can be defined by $x_n = f(x_{n-1})$. Many of them undergo an interesting transition from a stable state to a chaotic one as some variable in the function is increased. For some values of the variable, they converge to a stable fixed point. As the variable increases, they converge to a cycle of 2 values, then 4 values, 8 values, etc — a phenomenon called period bifurcation (fig. 1). At some value, the cycle becomes infinitely large and the system becomes chaotic. In 1970's, many physicists studied systems of fluid dynamics. Such systems were modeled using very complicated dynamical systems, which were practically impossible to analyze, especially with computers of that time. Feigenbaum, however, approached the problem differently. He claimed that the general behavior is the same for *all* bifurcating systems. Thus, he started working with less complicated ones instead. The simplest dynamical system that bifurcates is $x_n = kx_{n-1}(1 - x_{n-1})$ where x_n is the n th system value. For values of $k < 3$, the system converges to a stable point. When $k > 3$, the system converges to a cycle of 2 values (also called a 2-cycle) and does the same for all values until 3.45. At 3.45, the 2-cycle becomes a 4-cycle, and that becomes an 8-cycle at 3.54. The bifurcation continues until about 3.57, when the cycle becomes infinitely large and the system becomes chaotic. For any positive integer n , we can find a corresponding value A_n at which the 2^n -cycle splits into the 2^{n+1} -cycle. Feigenbaum analyzed the sequence of these values. He found that the distances between them decreased by a factor that approached a certain number as the cycle size approached infinity. What was even more amazing is that later, when analyzing a totally different system, $x_n = k \sin x_{n-1}$, he found the same phenomenon with the same number. That number became known as the Feigenbaum's constant.

$$F = \lim_{n \rightarrow \infty} \frac{A_n - A_{n+1}}{A_{n+1} - A_{n+2}} \approx 4.66920160910299...$$

where A_n is the n th bifurcation value. Just as π , the constant is irrational and most recently was calculated up to about 500 digits after the decimal point (see [2]). After encountering the same number in different systems, Feigenbaum proved using a method called renormalization, that it was indeed universal for all bifurcating systems.

This discovery implied that the same phenomenon should be found in physical systems, including the old systems of fluid dynamics. It is well known that the motion of fluid molecules is chaotic. However, at the absolute zero (-273°C) there is no movement at all. Libchaber studied the transition from this stable state to a chaotic one using liquid helium. He found that as the helium was heated, atoms started vibrating periodically. At certain temperatures, the period doubled and at -269°C normal chaotic behavior began. After Feigenbaum's discovery, Libchaber measured the ratio of successive temperature differences. It turned out to be about 4.669. Shortly after, experiments like that

were done in practically every area of science. Feigenbaum's constant was found in all of the tested systems. It was also found in fractals, which exhibit chaotic patterns. For example, in the famous Mandelbrot set (fig. 2), there is a series of circular figures on the x-axis, which decrease towards the left of the fractal. The diameter of each one is about 4.669 times bigger than the one to the left of it.

[1] Favorite Mathematical Constants. Online.

(<http://www.mathsoft.com/asolve/constant/constant.html>)

[2] Feigenbaum's Constant Approximation. Online.

(<http://www.lacim.uqam.ca/piDATA/feigenbaum.html>)

[3] Lauweriew, Hendrik A. Fractals: Endlessly Repeated Geometrical Figures. trans. Sophia Gill-Hoffstadt. Princeton: Princeton UP, 1991.

[4] Steward, Ian. Does God Play Dice?: The Mathematics of Chaos. New York: Blackwell, 1989.

Knot Addition

Artem Ivashchenko

At first glance one would think that there is absolutely nothing special about knots. They were developed just to keep things secured and tightened. However, I chose the following topic because it represents a deep connection between knots and numbers. Knots can be added together just like numbers to form another knot.

In general, a knot is a closed loop in three-dimensional space, the ends of which are tied together, two distinct points of which never coincide. Let us return to the representation of knots on a 2D plane. If a knot is in regular position, it is given an ambiguous representation of knots. In order to avoid trivialities, one has to specify at a double point which line segment goes over and which passes under in accordance with its natural positions in 3D space. This is done using the following convention:

Knot addition can be done as described below. One removes of a small segment from each knot and connects the loose ends such that no new crossings are introduced and the result is a single knot.



The result is called the *connected sum*. The following picture gives an example of knot addition.

One way to think of knot addition is by comparing it to multiplication of natural numbers. Here is where we meet the concepts *knot* and *prime knot*.

Zero knot (unknot) is a knot with no crossings. Also called the trivial knot, equivalent to a plain circle. The zero knot like the additive identity in the adding of knots.

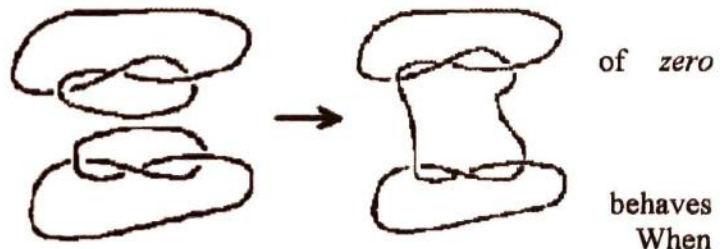
thinking knot addition, the zero knot plays a role very much like the number 1 plays when one thinks about prime numbers and multiplication. When one adds the zero knot to another knot, the knot is not changed. When one multiplies a number by 1, the result is always the same number. If you remove, or subtract the zero knot from a knot, it is not changed. Similarly, dividing a number by 1 leaves that number unchanged.

One can look at a knot and ask, "Are there 2 simpler knots that can be added together to produce this knot?" If the answer is "no", then the knot is a *prime knot*.

Prime Knot is a knot that cannot be broken down into a connected sum of knots, other than itself and an unknot. A knot that is not prime is called a *composite knot*.

Instantly, an association with prime and composite numbers arises. Multiplying together different combinations of prime numbers can produce all numbers that are not prime. The same thing is true with knots. Adding a finite union of prime knots can produce any composite knot. Indeed, there are a lot of prime knots that exist for knot addition.

As numbers, knots can also double up. The result is called a *link*. Link is a finite union of disjoint polygonal knots in 3D space. Links are represented on the 2D plane just as knots are, since they can be



made up of any number of connected knots. The various disjoint knots, which make up a link, are called components of a link.

Perhaps the most interesting links are *Borromean rings*. If you look at the links you will notice, that even though none of the components is connected to any other one, they still are connected in a link.

This gives you some idea why, although the concept of a knot is very simple mathematically, the territory for research is enormously vast.



It's Behind Door Number n

Brandon Irizarry

Ever heard of the Monty Hall problem?

"Suppose you're on a game show, and you're given a choice of three doors. Behind one door lies the grand prize; behind the others, goats. You pick door No. 1, and the host (Monty Hall), who knows what's behind the doors, opens another door, No. 3, which turns out to have a goat. He then says to you, 'Do you want to switch to Door No. 2?' Is it to your advantage in any way to say yes? Why?"

The answer that almost everyone gives to this problem is that it doesn't make a difference, for whatever reason. When this problem appeared in an issue of Parade magazine with the correct solution, even academics went into an uproar over how that solution was so obviously wrong. What is the solution? First, it's worth noting that the host could've opened Door No. 2 and given you the option of switching to Door No. 3; or, that you could've started out with any of the three doors, and the host would proceed in a similar manner. In fact, the first step towards understanding this problem is to realize that there are three possibilities, or cases involved, which can be systematically tabulated in a strategy matrix, as shown below:

case	Door		
	1	2	3
1	goat	goat	prize
2	goat	prize	goat
3	prize	goat	goat

As the contestant, you're keeping in mind three cases, or possibilities for the content behind each door.

From this matrix, and in general, it's evident that there's a $1/3$ chance of picking a prize door. So your first action, picking Door No. 1, has a $1/3$ chance of being successful. Now, the host knows what's behind each of the doors. Furthermore, if he's going to open a door for you, he certainly won't open the one with the prize. The second action is when the host opens a goat door. How does that change the strategy matrix?

Examining it case by case:

case 1	Door		
	1	2	3
case 1	goat	N/A	prize

The host has opened the remaining goat door. If you decide to switch, you'll win the prize.

case 2	Door		
	1	2	3
case 2	goat	prize	N/A

Same situation: if you decide to switch, you'll win the prize.

	Door	1	2	3
case 3	prize	goat	goat	

Either Door No. 2 or 3 may be excluded as inapplicable choices. Of course, which door is excluded doesn't affect your chances of success in any way, because if you choose either door, you'll lose. That also means that, if you decide to switch, you won't win the prize.

In light of this analysis, we see that switching leads to a favorable result in two of the three cases, which can only mean that the probability of winning the prize without switching is $1/3$ (the same probability of winning that you had just from choosing the first door), and after switching is $2/3$. Therefore, it must be to your advantage to switch, since your chances of winning go from $1/3$ to $2/3$.

This problem can be generalized for any number of doors; that is, it can be shown that, for any number of doors, it's always good to switch. (Of course, if there are one or two doors, there's no point to the game.) How about for four doors? Assume you always choose the first door.

	Door				
	1	2	3	4	
case					
1	goat	goat	goat	prize	
2	goat	goat	prize	goat	
3	goat	prize	goat	goat	
4	prize	goat	goat	goat	

Each case is examined like the three door situation. Here, in cases 1 to 3, switching gives you a $1/2$ chance of success (within each case), since there'll be a goat and the prize left (try it!). In case 4 you lose if you switch. There are four cases, which means our probabilities will be of the form $k/4$, where k is a rational number. We can see immediately that the probability of winning the prize by sticking to the first choice is $1/4$. By switching, our chances of winning become the sum of the chances of success within each case:

$$(1/2)/4 + (1/2)/4 + (1/2)/4 + 0 = 3/8$$

This is greater than $1/4$, which means that it is to our advantage to switch.

One more question. For any number of doors, will it always be to our advantage to switch? Let's see. We would have n doors, and thus n cases, resulting in an $n \times n$ strategy matrix. For cases 1 to $n - 1$, by sticking to our first choice, we'd lose; for case n , we'd win. That has a probability of $1/n$. For all cases, remember that the host will always eliminate only one losing door. Therefore, for cases 1 to $n - 1$, switching leaves us with a $1/(n - 2)$ chance of winning within each case; to find the general probability of success by switching, we have:

$$1/(n(n - 2)) + 1/(n(n - 2)) + 1/(n(n - 2)) + \dots 1/(n(n - 2)) + 0$$

where $1/(n(n-2))$ is taken $n-1$ times, the probability of winning by switching in case n being 0. Thus, the general expression for the probability of winning by answering "yes" to the host's offer is $(n-1)/(n(n-2))$.

Now, for n equal or greater than 3, is that expression always greater than $1/n$? Of course, since if

$$1/n < (n-1)/(n(n-2))$$

then

$$(n-2)/(n(n-2)) < (n-1)/(n(n-2))$$

$$n-2 < n-1$$

$$n < n+1$$

which is very evidently true. (Note that the original inequality is undefined when $n=0$ or when $n=2$. The case $n=1$ yields the absurdity $1 < 0$.) In conclusion, we've seen that, for any number of doors, it's advantageous to switch.

In writing this article, I had the help of the following sites: www.intergalact.com/threedoor/ndoor.html,

and

www.comedia.com/hot/monty-answer.html. Of course, it was up to me to make an article out of this extremely sparse amount of material. ;-)

Networks

Pamela Yang

When you were in grade school, do you recall classmates asking you to draw certain figures without lifting your pencil and without going over a line more than once? There was this one particular example that puzzled me for over a week, but at the time I was only eight. It looked something like figure 1.

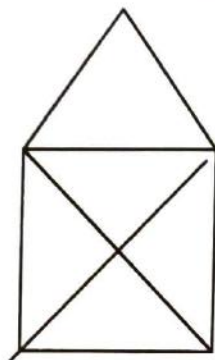


Fig. 1

Some diagrams similar to this may be possible to draw under these conditions and others that are impossible. These figures are called networks, a configuration made up of line segments and/or continuous arcs. There are certain terms that are used when discussing networks.

In any configuration, the point at which the line segments and/or arcs intersect is called the vertices. In networks, we can refer to them as *nodes*. The set of line segments and/or arcs that connect certain pairs of nodes are called the *edges*. The *degree* of a vertex is the number of line segments and/or arcs that have an endpoint at a particular node. For example, in figure two, line segments AB and DB, and arc DB all have an endpoint at B. Therefore, node B has an odd degree of three.

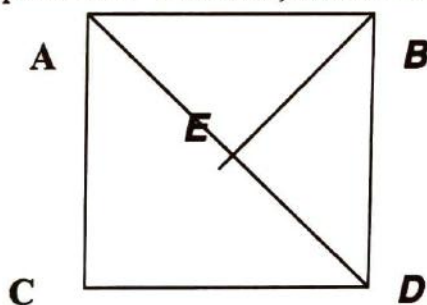


Fig. 2

There are two types of networks, one that can be traversed, or traced, and one that cannot. One mathematician who worked with these such networks named Leonhard Euler (pronounced "oiler"), discovered two types of traversible networks. One is called an Euler circuit, the other an Euler path. Euler circuits are networks that begin and end in the same point or vertex and goes through each edge only one time. An Euler path basically has the same definition, except it may end at a different point than from which it started.

There are two basic theorems that are used in determining the classification of a network:

1. A network contains an Euler circuit if and only if it has NO vertices of odd degree. It contains a Euler path if and only if it has either zero or two odd degree vertices.²
2. In a traversible network, there must be an even number of odd degree vertices.³

Using these two theorems we can easily decide whether certain figures contain Euler circuits or Euler paths, whether we are able to trace them or not, without going crazy trying to replicate them.

The most famous and popular problem concerning networks and Euler circuits and paths was the "Seven Bridges of Konigsberg." The problem goes something like this:

During the eighteenth century, the Pregel River divided the city of Konigsberg (located in East Prussia) into four sections, which included the island of Kneiphof. Seven bridges were built to connect the four regions. It was said that the people of Konigsberg spent their Sundays walking around, trying to find a starting point so that they could walk about the city, cross each bridge once and only once, and then return to their starting point.⁴

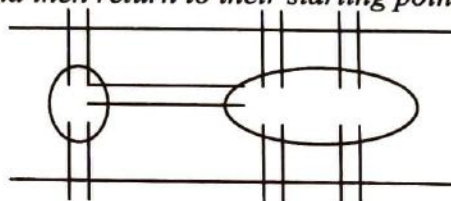


Fig. 5

How do you go about solving this problem? Euler said it couldn't be solved. Well, you can treat it like a network. By labeling the island of Kneiphof "A", the region on the bottom "B", the region on top "C", and the other island on the right "D" you can use these as the nodes. By connecting all of these you can form a network. The resulting network looks something like this.

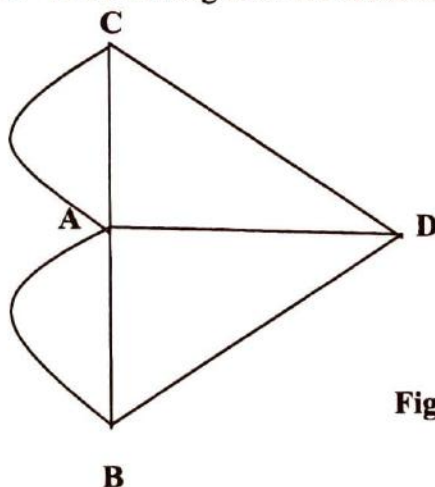


Fig. 6

From this figure you can find the solution. "A" has a degree of five, "B" has a degree of three, "C" also has a degree of three, and "D" again has a degree of three. Since there are four nodes and all of them have odd degrees we are not able to traverse this network. I guess Euler was correct in saying that this could not be accomplished. I bet you never thought that networks could be in any way, shape or form related to math. For those people who are paperboys, deliverymen/women, mailmen/women, street cleaners and garbage men/women getting the job done quickly and efficiently is very important. Retracing routes would only be extra work on their part. By applying networks to their routes maybe they can find an easier way.

The Four Color Problem

Anand Omprakash

The four-color conjecture is one of the most intriguing puzzles in all of mathematics that can be fully understood by non-mathematicians. Although a proof for it was created in the 1970's (Appel and Haken), it used a substantial amount of computer work and could not completely be reviewed by hand because of the massive amount of information that it processed by computer.

The problem asks whether or not any map with a significant number of regions can be colored with a maximum number of four colors such that no two adjacent countries or regions share a common boundary. A proof was formulated for any map with fewer than 25 regions.

The problem dates back to ancient Greece where it was first developed. A Greek mathematician stated that four colors suffice to color any map. He did not leave a proof of the problem, and the conjecture went unknown for nearly two thousand years. In the mid 1800's, Francis Guthrie displayed the problem to a number of English mathematicians, all of whom attempted to solve the conjecture in vain.

One of the most famous flawed proofs produced for the conjecture was by an English mathematician by the name of Kempe. The error in Kempe's proof went unknown for 11 years when Heawood found an error in it. While Kempe was not able to solve the problem, the information and concepts he used were salvaged to make a proof for the Five Color Theorem and also to show that no five countries can be mutually adjacent, which became a main element of several flawed proofs created thereafter.

In order to be able to approach the four-color problem much more intuitively, it is necessary to relate the problem to graph theory. For any given map establish an individual node in each of the regions. This node will represent the country. Where two countries are adjacent by a boundary, connect the two countries' respective nodes. Eventually the map will represent a graph. This sort of graph is referred to as a dual graph $D(G)$, because it is essentially dual or parallel to a regular map.

A graph of this sort can also be referred to as the underlying graph of M , or $G = U(M)$. The underlying graph refers to any graph that can represent a single map.

In graphs, the degree of a valence, or vertex, refers to how many edges collide with it. It can be denoted by $\deg(v)$, where v is the vertex. A vertex with the least number of points colliding with it is $\delta(G)$, while that vertex which is incident with the most number of vertices in a graph is $\Delta(G)$. A loop is an edge with only one endpoint, and parallel edges are edges that connect the same two countries. For any map, the property of being k -connected refers to being able to lose any set of $k-1$ points and still be connected.

A planar graph is one in which no two edges meet at any area other than an endpoint. All graphs pertaining to the four-color problem are planar and are one in the same as maps, or dual. A circuit is a continuous chain of points (regions) that form a complete chain. The removal of any edge in such a chain disrupts the circuit. In graphs, a tree refers to a connected graph with no circuits, and a tree edge is called a branch. Bridges are links that do not lie in circuits. In addition, when a map or planar map can be colored using k amount of colors, it is said to be k -colored.

In dual graphs, there is a one to one correspondence between the map region and its vertex. This correspondence can be written as follows. Let M be a map and let $D(M)$ be the dual graph of map M .

The correspondence between regions and vertices can be stated as $R \rightarrow R^*$. If R and S are regions of M , one can draw an edge e for every edge that lies in the common boundary. The edges e and e^* are considered to be orthogonal to one another. It is easy to see that any graph $D(M)$ will be planar because there can be no

areas or points where two boundaries exist.

A chain of countries is said to be a continuous linkage of countries that can form a chain, or even a circuit. Kempe used this method to create his famous flawed proof. A cubic map is one where each vertex has a degree of 3.

Euler's formula relates the numerical properties of the numbers of vertices, edges, and regions, respectively stated symbolically as V , E , and R . The formula states that in any planar graph, $V - E + R$ is always equal to 1. This condition is a necessary element in every planar map because it relates the all three of these characteristics in such a way that one can always determine whether or not the graph is planar.

In a tree, $V - E = 1$. In addition, the value of R is 0. To show this, let R and $E = 0$. Therefore, the graph consists of a single point V , and so the formula holds. As a result, $V - E + R = 1$.

The chromatic number refers to the minimum number of color required to color any map. The chromatic number of any map or graph can be represented by $\chi(G)$. In addition, one may also state that $\chi(G) \leq 1 + \Delta(G)$, where $\Delta(G)$ denotes the maximum degree for any vertex, and $\chi(G)$ denotes the chromatic number of G . The preceding formula merely gives a bound for the chromatic number. The proof of it is fairly simple.

By induction, one can figure out that any graph that can be k -colored must have at least one point that has a degree value less than or equal to k . Assume that this formula holds for G if G has $n-1$ vertices, and let G be any graph with n vertices. This allows us to conclude that $\chi(G') \leq 1 + \Delta(G') \leq 1 + \Delta(G)$ when (G') is the graph of G without its maximum degree vertex, which can be represented by v . Therefore, v can be colored with the last color that is not adjacent to v .

The Heawood number refers to the bound on the number of colors required to color any surface except for a plane. In addition, it was proven to also be the number of colors necessary to color a surface.

The Heawood formula can be used to calculate the chromatic number of any surface except for the plane. This is so because the Euler Characteristic for the plane is 2, and the maximum value of n that can be used is one. The Heawood number for any surface except for a plane can be calculated using the following equation:

$$\frac{1}{2}\{7 + \sqrt{49 - 24n}\}$$

where n is equal to the Euler Characteristic for any surface.

There is much more to the Four-Color Problem than just four-coloring maps. The Four-Color Problem is a study of graph theory in its most basic form. Studying the conjecture also leads one to other topics in coloring such as the Heawood Number, and Euler's Formula. These different topics relate to the Four Color Problem in that they both provide basic concepts as to how to approach and discuss the problem.

Al-Khwarizmi and a History of Algebra

Tayeb-Al-Karim

We use algebra almost every day in class. Our math teacher, Mr. Clancy, makes us do the stuff every day as a "Do Now".

The biographer Haji Khalfa states that al-Khwarizmi was the first Islamic author to write, "on the solution of problems by *al-jabr* and *al-muqabala*" (you can probably guess that the word *algebra* comes from the word *al-jabr*). The usual meaning of *jabr* in math is: adding equal terms to both sides of an equation in order to eliminate negative terms. Another, less used, meaning is: multiplying both sides of the equation by one and the same number in order to eliminate fractions. *Al-muqabala* usually means the "reduction of positive terms by subtracting equal amounts from both sides of an equation. But al-Karaji used it in the sense: to equate.

The combination of the two words, forming the word *al-jabr wal-muqabala* is sometimes thought of in the general sense of its meaning: performing algebraic operations. This is fine, but it can also be taken to mean the science of algebra.

The following is an example of the use of these words in the work of al-Khwarizmi. In Rosen's translation of the "Algebra of Mohammed ben Musa", the following question is posed:

I have divided ten into two portions. I have multiplied the one of the two portions by the other. After this, I have multiplied one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other.

Al-Khwarizmi calls one of these portions "thing" and the other, "ten minus thing". Multiplying by two, he obtains, in Rosen's translation, "ten things minus a square". In this way, he gets the equation, "A square, which is equal to forty things minus four squares".

In modern notation, we write the equation this way: $x^2 = 40x - 4x^2$.

Next, after using the operation *al-jabr*, al-Khwarizmi adds $4x^2$ to both sides, in this way obtaining: $5x^2 = 40x$ or $x^2 = 8x$. From here, he obtains: $x = 8$.

As another example, al-Khwarizmi has the equation: $50x + x^2 = 29 + 10x$, which is reduced by *al-maqabala* to: $21 + x^2 = 10x$.

In the first part of al-Khwarizmi's treatise entitled, "The Compendious Book on Calculation by *al-jabr and al-muqabala*", he explains the solution of six types, to which all linear and quadratic equations could be reduced:

- | | |
|-----------------|---------------------|
| (1) $ax^2 = bx$ | (4) $ax^2 + bx = c$ |
| (2) $ax^2 = b$ | (5) $ax^2 + c = bx$ |
| (3) $ax = b$ | (6) $ax^2 = bx + c$ |

where a , b , and c are all given positive numbers.

The third and largest part of the algebra of al-Khwarizmi deals with legacies. It consists entirely of problems with solutions. The solutions involve only simple arithmetic or linear equations, but they require considerable understanding of the Islamic law of inheritance.

Pascal's Triangle

Timothy Young

Pascal's triangle is a triangle that contains many unique properties and patterns. It is constructed by starting the top two rows as 1 and 1 1. In order to construct the following rows, add the two numbers that are directly to the right and to the left of it, placing a 1 at both ends of the row. For example, the first five rows of Pascal's triangle would look like this:

Row 0	1
Row 1	1 1
Row 2	1 2 1
Row 3	1 3 3 1
Row 4	1 4 6 4 1
Row 5	1 5 10 10 5 1
Row 6	1 6 15 20 15 6 1

As the triangle expands, many different patterns can be seen. For example, if you take the sum of each row, you can see that each row is equal to a power of two. Since so many exist, it is extremely difficult to find every one of them. Many made attempts in solving the mystery behind this arrangement of numbers, ranging back to the 12th century. The triangle was named after Pascal after he had published his findings on the triangle.

There are many different number patterns that can be found in Pascal's triangle. One of the patterns is the sequence of the powers of 2. This sequence is 1, 2, 4, 8, 16, 32, 64, 128 and so on. This sequence can be found if you add the sum of each row and match the row number to a power of 2. For instance, in the second row, the numbers are: 1 2 1. These three numbers add up to four or 2^2 . The third row of numbers is 1 3 3 1, or 2^3 .

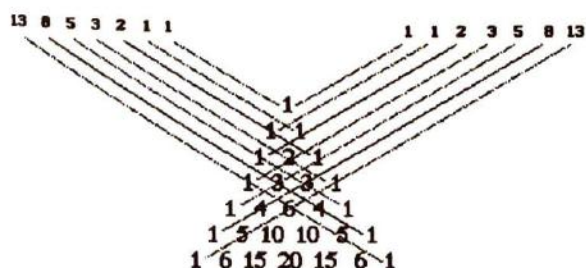
$$2^{\text{nd}} \text{ Row: } 1 + 2 + 1 = 4$$

$$3^{\text{rd}} \text{ Row: } 1 + 3 + 3 + 1 = 8$$

$$4^{\text{th}} \text{ Row: } 1 + 4 + 6 + 4 + 1 = 16$$

Another particular sequence that can be found in Pascal's triangle is the fibonacci sequence. The fibonacci sequence is a sequence that is found by starting with two numbers. The last two numbers of the sequence are then added together to produce the next number in the sequence. For example, if we start the sequence with two one's, this is how the sequence would look like:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 477, 710, 1187,



In order to find the Fibonacci sequence within Pascal's triangle, you must look at the "rising diagonals", adding the numbers found when lines

Many other types of number patterns can also be found in Pascal's Triangle. For Instance, triangular numbers can be found. Triangular numbers are figurate numbers, numbers that are represented by geometric patterns created by equally spacing points. By following the third diagonal, the numbers appear.

0 th row	1									
1 st row		1								1
2 nd row			1			1				2
3 rd row		1			1				3	
4 th row	3		1							
5 th row	6		4	1			1		4	
6 th row	10	1			5			1		10
7 th row			5				1			

Pascal's triangle is a very unique and interesting figure. Many different patterns can be extracted from this arrangement of numbers with many still not found yet.

Fibonacci Numbers

Daniel Geeng

In the thirteenth century in Italy lived a great expert and calculation and numbers. His name was Leonardo (" of Pisa", as he was called), he was also known as Fibonacci.

In 1202, he published a book in Latin, Liber Abaci (the book about Abaci), which contained all that was the known about arithmetic and algebra. It was one of the first books in Europe to teach calculation with Arabic numerals. For more than two centuries it was the definitive work on numerical calculations. As was the custom, Fibonacci participated in mathematical tournaments (public contest for the best and speediest solutions of difficult problems). His skill in solving numerical problems was astonishing.

Fibonacci's biggest accomplishment is creating and working with the Fibonacci numbers:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89,...

Each number equals the sum of the preceding two: $1+1=2$, $1+2=3$, and so on.

If two consecutive numbers in the series are y and x , then either:

For example: $X=2, Y=1$;
 $X=8, Y=5$;
 $X=13, Y=8$;

Are roots of the first equation, and:

$X=3, Y=2$;
 $X=8, Y=5$;
 $X=21, Y=13$;

Here is a very interesting problem.

We all know that when you cut up a figure and reorder the parts, the shape may change but, clearly, the area cannot.

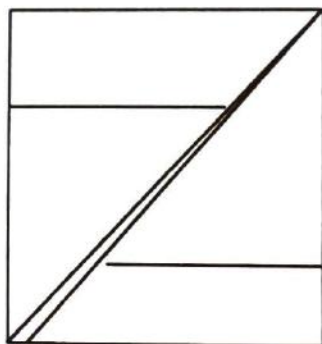
But consider the diagrams above. The square is cut into two congruent triangles and two congruent trapezoids. Can we choose x and y so that the square can transform into a rectangle as shown?

The result is confusing. Using $x=5$ and $y=3$, the rectangle's area is 65. But the area of the square is only 64!

When we try a 13- by 13 square($x=8, y=5$), the rectangle's area was 168 instead of 169. The same holds true for a 21- by - 21 square($x= 13, y=8$)

What is wrong here, and what does it have to do with Fibonacci numbers?(you should observe that the lengths for x and y are all consecutive Fibonacci numbers)

The result is not very complicated. The rectangle's area, $x(2x+y)$, minus the square's, $(x+y)^2 = x^2 - xy - y^2$, which as we know from the properties of the Fibonacci series above equals to 1 or -1 when x and y are Fibonacci numbers, as here. The rectangle's parts overlap slightly or do not quite meet, which accounts for the difference in the area. The diagram shows the " hole" KHEF in the 13-by - 5 rectangle, a thin quadrilateral with an area of 1.



To prove it, let M be the point where the extension of side EF of triangle EFN crosses KL . (If EFK is a straight line, M coincides with K .)

Since triangle EFN and EML are similar , we have these ratios:

$$\frac{ML}{FN} = \frac{EL}{EN}, \text{ or } \frac{ML}{FN} = \frac{13}{8}$$

Then $ML= 4.875$. Since $KL = 5$, M does not coincide with k . Therefore, EFK and EHK are not straight lines, and there is a hole shown.

To cut the square into a true rectangle, we let $x^2 - xy - y^2 = 0$ instead of 1 or -1 to erase that hole. Solving, and accepting only the positive solution:

This is the ratio of "golden sections," long valued by architects and scientists.

Here is a little extra. Can you show that the sum of the first ten terms of both the Fibonacci and the Fibonacci-like sequence equals 11 times the 7th term in that sequence?

Let the first number be a , the second be b .

Term	Fibonacci-like sequence
1	A
2	B
3	A + B
4	A + 2B
5	2A + 3B
6	3A + 5B
7	5A + 8B
8	8A + 13B
9	13A + 21B
10	21A + 34B

$$\text{Sum} = 55a + 88b = 11(5a + 8b)$$

Challenge: Describe a way to find the sum of the first n Fibonacci numbers.

First, study the table below and try to find a pattern on the first ten numbers of the Fibonacci numbers.

n	Sum of the 1 st n Fibonacci numbers	It's found by subtracting	From the Fibonacci number
2	2	1	3
3	4	1	5
4	7	1	8
5	12	1	13
6	20	1	21
7	33	1	34
8	54	1	55
9	88	1	89
10	143	1	144

The sum of the first n Fibonacci numbers is found by

The Fibonacci Quarterly is the official journal of the Fibonacci association. This quarterly is a rich source of information on Fibonacci numbers. It is highly recommended for additional information on the Fibonacci numbers.

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Bridge Geometries

Cindy Lin

Every year students in research physics have to build a bridge from 15 sticks of 3/32 inch square cross-section basswood. The bridge has to span a gap of 30 cm but be less than 40 cm. It can be at most 12 cm high and 8 cm wide. It can weigh at most 25 grams. It has to be constructed to provide support for at three loading points: one at the center, one 4 cm to the right of the center, and one 6 cm to the left of the center.

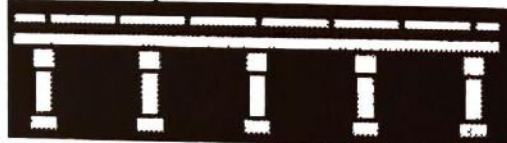
Testing is done by putting a 4 cm x 4 cm square plate that is at least 6 mm but less than 13 mm thick on one of the loading points. A 9.53 mm diameter eyebolt is attached to the loading plate. The loading plate and the eyebolt provide a minimum initial load of 2 kg. Additional load is then applied by hanging masses on the loading rod. The bridge is designated as a failure when is it no longer able to carry additional load or when it reaches a load deflation of 25 mm under the loading point, whichever occurs first. The winners are selected by using a formula to calculate the efficiency E:

$$E = \text{load supported in grams} / \text{mass of bridge in grams}$$

The winning bridge is the one with the highest efficiency.

Although this is an activity for a Physics class, the construction of the bridges requires good knowledge of geometry. There are five basic kinds of bridge geometries: beam, truss, arch, suspension, and cantilever.

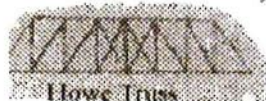
A. The "Beam Bridge" is a good for spanning a short gap that isn't very high. It is supported on both ends by land or tall columns. If a beam bridge needs be longer, support columns can be used to maintain support throughout the span. The beam has to withstand both compression in its upper parts and tension in its lower parts.



B. The "Truss Bridge" is similar to a beam bridge but much stronger. These bridges use a pattern of triangles called trusses. The trusses are very rigid and thus make the bridge very strong. Because of their strength, truss bridges are often used for railway bridges as they must be able to support the great mass and the vibrations of the trains.



Town Lattice Truss



Howe Truss



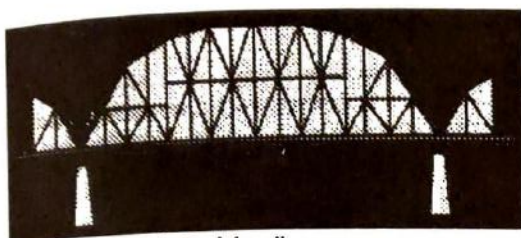
Pratt Truss



Baltimore Truss



Warren Truss

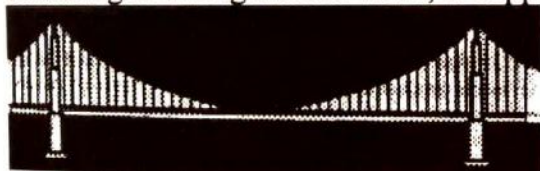


C. The "Arch Bridge" can span a very deep chasm and maintain support throughout the span. The arch bridge is strong because no one point along the arch supports more mass than any other point - thus, the weight, which is spread out both downwards and outwards, is spread out evenly throughout the bridge.

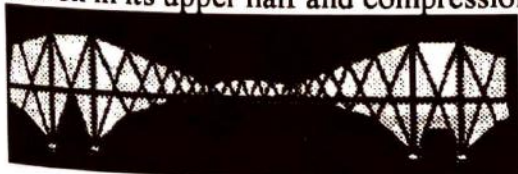
THE TWO SIDES SUPPORT THE WEIGHT IN AN OVERLAPPING DIAGONAL.



D. The "Suspension Bridge" is the best kind of bridge to build when trying to span a very long gap. It consists, basically, of a deck suspended from cables slung between high towers. The cables of high tensile steel wire can support an immense weight. The towers are in compression and the deck, often consisting of a long slender truss, is supported at frequent intervals along its length.



E. The "Cantilever Bridge" is constructed from two beams jutting out from either side of a riverbank or body of water (embankments). These two beams must be securely anchored into the embankments using concrete slabs called abutments. The two beams meet in the middle of the bridge. If the bridge needs to be longer, then a third beam is added between the first two and this third beam must be supported by a pair of columns or a center pier. Unlike a simple beam supported at both ends, the cantilever must resist tension in its upper half and compression in its lower.



At the following website: <http://www.dean.usma.edu/cme/civilsoft.htm>, there are six civil engineering programs that can be downloaded. One of them is the "The West Point Bridge Designer." The program introduces you to the engineering design process by guiding you through the design of a highway bridge structure. For beginners, there are templates that can be used. After the design is

finished, the program can check the stability of the bridge and correct errors, if any. When you have completed your design, the program will run a simulated load test of your bridge and compute its cost. This is a great program to learn more about bridge geometries and apply them first hand.

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Matrices, Probability and Game Theory

Benjamin Lerner

Game theory deals with mathematically analyzing situations in which there is a conflict of interests, and determining what course or courses of action will result in the best possible outcome. Although it can be used to analyze simple games such as card or board games, this barely scratches the surface of its potential. Game theory as a field can be broken down into a hierarchy of sub levels, each level simpler to analyze than the ones above it.

Game theory was first introduced by Émile Borel, in 1921, when he attempted to mathematically model previously recorded military engagements. However, the theory was firmly established by John von Neumann in 1928, when he proved the Minimax Theorem (discussed later), which has profound influence on the entire study of game theory. Game theory was brought to more public recognition when Neumann and Oskar Morgenstern published their groundbreaking work entitled Theory of Games and Economic Behavior. This work set forth the basic principles of game theory, and began the intense research in the subject, which continues through the present.

THE UNDERLYING TERMINOLOGY AND BACKGROUND:

Game theory can be broken down as a hierarchy of game types, with each level easier to analyze than the previous ones, but also much more specific. The first major subdivision separates the finite games from the infinite ones. In some cases, but not in general, the infinite games can be modeled by finite ones, or at least constraints can be found for them, so that they become tractable to analysis. The next major subdivision distinguishes between two-player and n-player games. Two player games are much easier to analyze than n-player games, because there is no possibility of cooperation between players, which immensely increases the number of possible courses of action. Two-player games can be broken down further into two-player constant sum games, and non-constant sum games.

To analyze any game, one needs to create a representation for that game, which expresses each players' choices for a move (known as a **strategy**), and each players' winnings (if any) for each move (known as the **payoff**). There are numerous to achieve this for n-player games, and since not all of them are equivalent, this leads to difficulties in analysis. However, to accomplish this for any two player game, one constructs a **payoff matrix**, an $m \times n$ matrix where m signifies all the strategies for the first player, known as the Row player, and n signifies all the strategies for the second, or Column, player. The entries in the matrix are the payoffs, ordered pairs in which the first element is the Row player's payoff and the second the Column's payoff. Each player moves simultaneously by choosing one of the available strategies, and such a move is called a **play** of the game. Naturally, this is a single-move game, and does not reflect many games familiar in everyday life, but single move games are useful in modeling real scenarios. Now that the underlying terminology is defined, we can turn our attention to analyzing games, in specific, finding the best strategy for either player in any given game.

TWO-PLAYER CONSTANT OR ZERO SUM GAMES

For any given constant sum game, we mean that when you add each player's payoff for any given move, it is a constant number. However, we can subtract that constant from every entry in the matrix, and the result is essentially the same game as before, since the relative maxima and minima are unchanged for each row and column, which means that the ideal strategies for each player are unchanged. Thus, instead of speaking about a constant sum game, we can instead speak about **zero-sum games**. Two person zero sum games are defined as two person games in which one player's loss is the

other player's gain. For example, take the game of Rock, Paper, Scissors. In that game, rock beats scissors, paper beats rock, and scissors beat paper. Now, to define the payoffs, let Row gain two dollars if he wins, Column gains one dollar if he wins, and each loses the same amount if he loses. So, the

payoff matrix looks like the following:
$$\begin{bmatrix} (0,0) & (-1,1) & (2,-2) \\ (2,-2) & (0,0) & (-1,1) \\ (-1,1) & (2,-2) & (0,0) \end{bmatrix}$$
. Notice that if we separate the

ordered pairs into two matrices,
$$\begin{bmatrix} 0 & -1 & 2 \\ 2 & 0 & -1 \\ -1 & 2 & 0 \end{bmatrix}$$
 and
$$\begin{bmatrix} 0 & 1 & -2 \\ -2 & 0 & 1 \\ 1 & -2 & 0 \end{bmatrix}$$
, each is the negative of the other.

So when speaking about zero-sum games, we do not require ordered pairs, but can simply write the payoffs for the Row player, and remember that if we want to find the payoffs for the column player, we negate the values.

Before we can begin to analyze what each player can do to maximize his personal payoff, we must make two assumptions - first, that the player is rational, that is, he will always act logically, and second, that his only concern is for himself. Now we are ready to begin. There are two types of strategies a player can choose - pure or mixed. A pure strategy means that the player will pick that strategy all the time, no exceptions. A mixed strategy means that the player will select at random from among a set of strategies, with weight given to better strategies. There are two methods of finding pure strategies: domination and saddle-point. Domination cannot always locate the pure strategy, so it mostly is used to ease the location of the saddle-point.

Domination can best be described as the elimination from consideration of all obviously poor choices. This can be done for the row player as follows: compare any two rows. If each entry in one row is greater than or equal to the corresponding entry in the other, then the second row is a poor choice, and is said to be **dominated** by the first. This argument is based on the assumption that each player is out to maximize his payoff - if one row is always less than the other, why choose it? To do this for the column player, remember that his payoffs are the negative of the ones in the payoff matrix, so he is looking for columns, which are less than the others, because to him they are better. The following

example demonstrates this idea of domination: Starting with the payoff matrix
$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 0 & 1 \\ 2 & 3 & -4 \end{bmatrix}$$
, one

can eliminate the first row, since each entry is smaller than the second row. Similarly, one can eliminate the second column, because the first one is smaller. This leaves us with the matrix
$$\begin{bmatrix} -1 & 1 \\ 2 & -4 \end{bmatrix}$$
.

However, this method cannot always be used, because sometimes no row is dominated by any other, or the matrix does not reduce to a one by one matrix. There is another method, called location of the saddle point, which invariably finds the best strategy for each player. This is done as follows: since the players are both rational, not only do they want to win as much as possible, but they want to cut their losses as much as possible. So, finding a saddle point consists of finding the best possible alternative in the worst case scenarios. This is done by finding the column maximum of the row minima. For

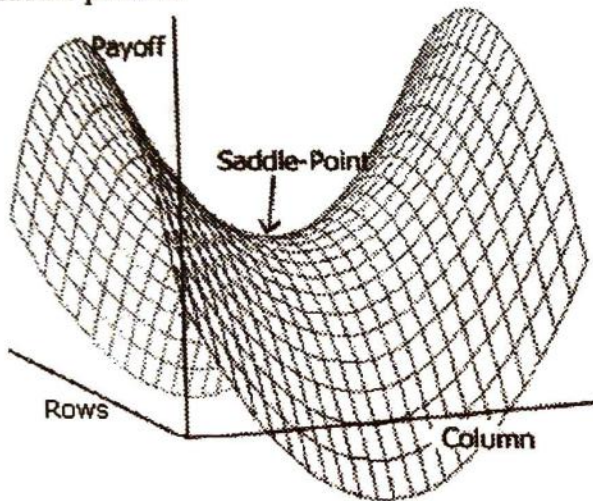
example, in the following game,
$$\begin{bmatrix} 0 & 5 & 10 \\ -5 & 0 & 10 \\ -10 & -10 & 0 \end{bmatrix}$$
, if we label the row minima we get

$$\begin{bmatrix} 0m & 5 & 10 \\ -5m & 0 & 10 \\ -10m & -10m & 0 \end{bmatrix}$$
 Notice there can be more than one row minimum, however, they are all equal.

Now we label each column maximum as well, giving the following matrix:

$$\begin{bmatrix} 0M & 5M & 10M \\ -5 & 0 & 10M \\ -10 & -10 & 0 \end{bmatrix}$$

Again, notice that there can be more than one row maximum, however, they are all equal. If any entry has an m(minimum) and an M(maximum), then we have found a **saddle point**, or a **minimax point**. A saddle point is



defined as any entry in the payoff matrix, such that any other choice by Row would leave him vulnerable to gaining less, and any other choice by Column would leave him vulnerable to the same. So the saddle point is the ideal solution for each player - it finds their pure strategies. In this example, each player's pure strategy would be strategy 1. If such a saddle point exists, then the game is said to be **strictly determined**. The value of the saddle point is said to be the **value** of the game, since, when the game is played, that outcome will always occur. Intuitively, if the value of the game is not zero, then the game is unfair, as one player can expect to win consistently.

There are times when there is no saddle point. In such a situation, there are no pure strategies for either player. (There cannot be a pure strategy for a single player without the other player having one also - if Column had a pure strategy j , then Row could ignore all the other columns, and then pick the row i whose entry i,j was a maximum for column j , thus giving him a pure strategy. The same argument holds for Row.) What is now required is to find the optimal mixed strategy for each player so that each player can maximize his *average* payoff.

This is where the power of matrices becomes very useful. We start with the payoff matrix, which will be denoted A . Next, we have two probability vectors: an m by 1 matrix, p , whose entries $i,1$ are the probabilities that Row will chose the corresponding strategy i , and an n by 1 matrix, q , whose entries are the probabilities that Column will chose the corresponding strategy. If we pre-multiply A by p^t and post-multiply it by q , or $p^t A q$, we obtain a single number which is the **expected value** of the game (for those two mixed strategies), denoted E . The reasoning is as follows: if we consider the last two terms, we get an m by 1 matrix whose entries, loosely speaking, are the expected payoffs to Row for each strategy he chooses. Now, if we multiply each of those values by the probability that Row will receive that payoff, we get the expected value. Note that when p and q are unit vectors (i.e.: all entries are 0 except a single 1), $p^t A q$ results in the number we previously defined as the value of the game. So the expected value is a generalization of the value of the game. If the expected value of the game is 0 , then we can again say that the game is fair, as overall each player can expect to win as often as he loses.

Now the question is how to find the ideal p and q so that Row maximizes E , and q minimizes it. For a two by two payoff matrix, this is easy. What we can do is, instead of having p and q be single

column vectors, let them be multi-column matrices, where each column represents a single mixed strategy, and further, let the probabilities change regularly, as in the following example:

$\begin{pmatrix} 1 & .8 & .6 & .5 & .4 & .2 & 0 \\ 0 & .2 & .4 & .5 & .6 & .8 & 1 \end{pmatrix}$ (the .5 disrupts the pattern, but it is sometimes helpful to leave it in, as the following example will show). Now when we find $p^t A q$, we find that the rows and columns correspond to the mixed strategies we used. Now we have a larger payoff matrix, but it has a very important property, which is illustrated in the following payoff matrix obtained from $\begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$:

$$\begin{pmatrix} 1 & .6 & .2 & 0 & -.2 & -.6 & -1 \\ .4 & .24 & .08 & 0 & -.08 & -.24 & -.4 \\ -.2 & -.12 & -.04 & 0 & .04 & .12 & .2 \\ -.5 & -.3 & -.1 & 0 & .1 & .3 & .5 \\ -.8 & -.48 & -.16 & 0 & .16 & .48 & .8 \\ -.14 & -.84 & -.28 & 0 & .28 & .84 & 1.4 \\ -.2 & -.12 & -.4 & 0 & .4 & 1.2 & 2 \end{pmatrix}$$

Somewhere in the matrix, there is an entire row and

column whose value is constant. In this matrix, only the column is apparent; it corresponds to a mixed strategy of (0.5, 0.5). We can find the location of the constant row by looking again at the maxima and

minima of the rows and columns:

$$\begin{pmatrix} 1M & .6M & .2M & 0M & -.2 & -.6 & -1m \\ .4 & .24 & .08 & 0M & -.08 & -.24 & -.4m \\ -.2m & -.12 & -.04 & 0M & .04 & .12 & .2 \\ -.5m & -.3 & -.1 & 0M & .1 & .3 & .5 \\ -.8m & -.48 & -.16 & 0M & .16 & .48 & .8 \\ -.14m & -.84 & -.28 & 0M & .28 & .84 & 1.4 \\ -.2m & -.12 & -.4 & 0M & .4M & 1.2M & 2M \end{pmatrix}$$

Notice that for the

columns, the column maximum shifts from the top of the matrix to the bottom along the constant column. The row minimum does not, but this is because the constant row is not shown. To find the mixed strategy of the constant row, we have to notice two things: first - the expected payoffs change linearly both across and down the matrix with the probabilities, and second - at some point the row minimum switches sides. If we graph any two columns of the matrix (preferably the left and rightmost ones), using the top row of p as the x coordinate and the value of the matrix at that point as y , we have two lines which cross each other somewhere between 0 and 1. To derive equations for those lines, we use pure strategies for Column, and $(x, 1-x)$ for row, to obtain two equations in x . Equating them in this case yields $3x-2=2-3x$, $6x=4$, $x=2/3$. So at $(2/3, 1/3)$, the corresponding row in the larger matrix would be $3(2/3)-2=0$. Notice that now there is a minimax point. This is what we were looking for: a pair of mixed strategies, which simultaneously allows Row to maximize E , and allows Column to minimize E . This solution only works for two by two matrices. However, a more general method from linear programming known as the **simplex method** enables the simultaneous calculation of p and q .

The simplex method works as follows: given the payoff matrix, first "translate" it so that all payoffs are positive by adding a constant to all entries. Remember the constant - later it will be needed again. Next, convert it to a system of linear inequalities. For example, if the payoff matrix is

$\begin{pmatrix} 3 & 2 & -1 \\ -2 & 0 & 1 \end{pmatrix}$, translate it to $\begin{pmatrix} 6 & 5 & 2 \\ 1 & 3 & 4 \end{pmatrix}$ by adding 3. Next, convert to inequalities:

$6p_1 + 5p_2 + 2p_3 \geq v$
 $1p_1 + 3p_2 + 4p_3 \geq v$. Dividing everything by v , and adding "slack variables" to convert the inequalities

to equations results in:
 $6x_1 + 5x_2 + 2x_3 + S_1 = 1$
 $1x_1 + 3x_2 + 4x_3 + S_2 = 1$, where $x = q/v$. Now we need an objective function, or the function we are looking to maximize. This is $1/E$, denoted w , and it equals $x_1 + x_2 + x_3$. Now, we

write this as a matrix:

	x_1	x_2	x_3	S_1	S_2	
S_1	6	5	2	1	0	1
S_2	1	3	4	0	1	1
w	-1	-1	-1	0	0	0

Notice that w has been negated - this will be

apparent later. The rules for pivoting are as follows: select the column (called the incoming column) whose value in the w row is the most negative. If there is a tie, select the leftmost one. This means that the variable that row represents (in the top row) can be modified to raise the value. Now, select the row whose rightmost entry when divided by the entry in the incoming column is the smallest positive number. This variable can be changed also. Now, divide the row by the pivot entry (the intersection of the incoming column and the row just selected), and, as in Gauss-Jordan elimination, add multiples of that row to the others to make all other entries in the incoming column 0. Finally, re-label the row so its heading is that of the incoming column. Repeat this pivoting until all the entries in the bottom row are

non-negative. Pivoting for the example once yields:

	x_1	x_2	x_3	S_1	S_2	
x_1	1	5/6	2/6	1/6	0	1/6
S_2	0	13/6	22/6	-1/6	1	5/6
w	0	-1/6	-4/6	0	0	1/6

Once

again yields:

	x_1	x_2	x_3	S_1	S_2	
x_1	1	7/11	0	2/11	0	1/11
x_3	0	13/22	1	-1/22	6/22	5/22
w	0	5/22	0	3/22	2/11	7/22

and we are done. Now, since $v = 1/w - 3$

(remember, we translated the game, so now we have to translate back) $= 22/7 - 3 = 1/7$, we can now find not only q , but p as well. Remember from before $x_i = q_i/w$, so $q_1 = (1/11)/(7/22) = 2/7$, $q_3 = (5/22)/(7/22) = 5/7$, and since x_2 is not listed as a row, $q_2 = 0$. Now to find p , use the S variables in the columns. $p_1 = (3/22)/(7/22) = 3/7$, and $p_2 = (2/11)/(7/22) = 4/7$. And so we have solved this game.

This completes the analysis of two person zero sum games. Using these techniques, we can solve any two person zero sum game, and find the optimal strategies and the expected value of the game. The next level from here is, naturally, two person non zero sum games, and that topic is beyond the scope of this article

Applications in Pascal's Triangle

Jennifer Lee

Pascal's triangle is one of the most famous and astonishing number patterns in mathematics. In Pascal's triangle, integers are arranged in an arithmetical triangle made up of staggered rows of numbers. It contains numerous number patterns, but it also has some useful math applications.

The intriguing Pascal's triangle was discovered about a millennium earlier and is still considered one of the most notable number patterns ever constructed in mathematics. The triangle is named after Blaise Pascal, a French mathematician who was the first one to write about the patterns of the triangle in a treatise called the *Treatise of the Arithmetic Triangle*. It was a book written in 1653 describing the different patterns found in Pascal's Triangle. However, the triangle was discovered several hundred years before Blaise Pascal.

The triangle very first appeared in the works of the Chinese and the Arabs in the early 11th century. Omar Khayyam, an Arab mathematician was the next to discuss the Pascal's triangle. In 1303, Yang Hui, a Chinese mathematician was the first to display an illustration of the triangular number pattern in his book. In 1653, Blaise Pascal's work on the topic in his treatise was considered most important as well as influential. Thus, the pattern was named after him.

One of the most well known characteristics of the Pascal Triangle is the binomial coefficients. When a binomial is raised to a certain power, one can easily find the coefficients of each of the terms. They form the rows of the Pascal Triangle. So if the binomial is $(x + y)^n$, the coefficients are in the n th row of the Pascal Triangle. The Pascal Triangle is very useful especially when the binomial is raised to a high power, or under great expansions. For example, if one wants to find the coefficients of a binomial raised to the 11th power, it would be much easier by referring to the 11th row of the Pascal's Triangle.

For every binomial expansion $(x + y)^n$, there are $n + 1$ terms. Also, the sum of the exponents of each individual expanded term is equal to the number of that power, or the row number of the Pascal Triangle.

In the Pascal's Formula, $\binom{n}{r}$ is a binomial coefficient. The coefficients are shown boldfaced in the following equations. Ways to write all the possibilities of choosing combinations of either x or y from four factors are:

$$\begin{aligned}(x + y)^4 &= (x + y)(x + y)(x + y)(x + y) \\(x + y)^4 &= {}^4C_0x^4y^0 + {}^4C_1x^3y^1 + {}^4C_2x^2y^2 + {}^4C_3x^1y^3 + {}^4C_4x^0y^4 \\(x + y)^4 &= \binom{4}{0}x^4y^0 + \binom{4}{1}x^3y^1 + \binom{4}{2}x^2y^2 + \binom{4}{3}x^1y^3 + \binom{4}{4}x^0y^4 \\(x + y)^4 &= 1x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + 1y^4\end{aligned}$$

The boldfaced coefficients of each of these rows are identical to the numbers that make up the fourth row of the Pascal's Triangle.

$$\begin{array}{rcl}(x + y)^0 & = & 1 \\(x + y)^1 & = & 1x + 1y \\(x + y)^2 & = & 1x^2 + 2xy + 1y^2 \\(x + y)^3 & = & 1x^3 + 3x^2y + 3xy^2 + 1y^3 \\(x + y)^4 & = & 1x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + 1y^4 \\(x + y)^5 & = & 1x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + 1y^5 \\(x + y)^6 & = & 1x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + 1y^6\end{array}$$

Binomial expansions

From the Pascal's triangle, n simply means the row number, and r means the place in that row. For example, to get to the leftmost 3 in the third row, the combinations that'll do are RLL, LRL, and LLR. To reach the 6 in the next row, there are 2 moves to the right and 2 to the left. Here are the possibilities: LLRR, LRLR, LRRL, RLLR, RLRL, and RRLR. The other way to look at this is that it counts the number of ways to put the 2 Rs in the 4 available slots. Therefore, the Pascal Triangle can come into use in probability problems.

Row 0				1				
Row 1				1	1			
Row 2			1	2	1			
Row 3		1	3	3	1			
Row 4		1	4	6	4	1		
Row 5	1	5	10	10	5	1		
Row 6	1	6	15	20	15	6	1	
	<i>0th</i>	<i>1st</i>	<i>2nd</i>	<i>3rd</i>	<i>4th</i>			

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Symmetry and Conservation Laws of Physics

Jerry Moy Chow

Symmetry is fascinating to the human mind. We are attracted to symmetrical objects or patterns. An isosceles triangle is more symmetrical than an arbitrary triangle. An equilateral triangle is more symmetrical than an isosceles triangle. A circle has the highest symmetry of them all. What is symmetry for the geometrical figures? If a figure or a system is unchanged, or invariant under a certain operation or transformation, then it is symmetric with respect to that operation or transformation. The more the number of operations leaving the system invariant is the higher the symmetry of the system. A circle has the highest symmetry because the number of transformations that leave it invariant is the most.

How can the laws of physics be symmetrical? There are operations we can do to physical phenomena, and leave the result unchanged. For example, the translation, rotation, reversion of space and time, will leave the result of physical phenomena the same so that the laws of physics are symmetrical in space-time. Symmetry is more interesting and beautiful in quantum mechanics, because for each symmetry, there is a corresponding conservation law and vice versa. Symmetry in classical physics, as well as in quantum mechanics will be introduced separately.

Symmetry of the Laws of Physics in Classical Physics

A. Space translation operation:

The laws of mechanics for a particle are given by Newton's law in vector form,

$$\vec{F} = m \frac{d^2}{dt^2} \vec{r} \quad (1)$$

or a set of three scalar equations as follows:

$$F_x = m \frac{d^2}{dt^2} x \quad F_y = m \frac{d^2}{dt^2} y \quad F_z = m \frac{d^2}{dt^2} z \quad (2)$$

Space translation of a coordinate system can be shown in figure 1. The relations between the prime and unprimed system are as follows:

$$x' = x - a \quad y' = y - b \quad z' = z - c \quad (3)$$

$$F_x' = F_x \quad F_y' = F_y \quad F_z' = F_z \quad (4)$$

Substitute the above quantities into equation (2) and we obtain

$$F_x' = m \frac{d^2}{dt^2} (x' + a) = m \frac{d^2}{dt^2} x' \quad F_y' = m \frac{d^2}{dt^2} (y' + b) = m \frac{d^2}{dt^2} y' \quad F_z' = m \frac{d^2}{dt^2} (z' + c) = m \frac{d^2}{dt^2} z' \quad (5)$$

or in vector form

$$\vec{F}' = m \frac{d^2}{dt^2} \vec{r}' \quad (6)$$

Newton's law has the same form. Therefore, this law of physics is symmetrical under space translation operation.

B. Space Rotations:

In order to test if the laws of physics are invariant under space rotation, we consider two coordinate

systems having different angular orientations as shown in figure 2.

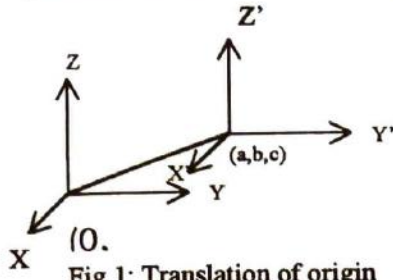


Fig 1: Translation of origin

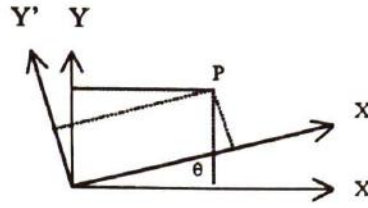


Fig 2: Rotation of the Coordinate System

The relationships of the quantities in equation (2) are as follows:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{or} \quad \begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= y \cos \theta - x \sin \theta \\ z' &= z \end{aligned} \quad (7)$$

$$\begin{pmatrix} F_x' \\ F_y' \\ F_z' \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix} \quad \text{or} \quad \begin{aligned} F_x' &= F_x \cos \theta + F_y \sin \theta \\ F_y' &= F_y \cos \theta - F_x \sin \theta \\ F_z' &= F_z \end{aligned} \quad (8)$$

Differentiate equation (7) twice with respect to t , and we have

$$\begin{aligned} m \frac{d^2 x'}{dt^2} &= m \frac{d^2 x}{dt^2} \cos \theta + m \frac{d^2 y}{dt^2} \sin \theta = F_x \cos \theta + F_y \sin \theta = F_x' \\ m \frac{d^2 y'}{dt^2} &= m \frac{d^2 y}{dt^2} \cos \theta - m \frac{d^2 x}{dt^2} \sin \theta = F_y \cos \theta - F_x \sin \theta = F_y' \\ m \frac{d^2 z'}{dt^2} &= m \frac{d^2 z}{dt^2} = F_z' \\ \text{or } m \frac{d^2 \vec{r}'}{dt^2} &= \vec{F}' \end{aligned} \quad (9)$$

Newton's law is invariant. Therefore, the laws of physics are symmetrical under space rotation operation.

Symmetry and Conservation Laws in Quantum Mechanics

In quantum mechanics the symmetry of physical systems under various operations and the principle of superposition of amplitudes are deeply related to conservation laws. The laws, which are symmetrical under space translation, space rotation, and time translation, turns out to be the conservation laws of momentum, angular momentum, and energy in quantum mechanics, respectively. There are many more symmetries in quantum mechanics, which have no classical analog, and have no method of description in classical physics. For example, the physical laws are unchanged if the phase of the wave function of a quantum mechanical system is shifted by a constant because the square of the wave function does not change. That is symmetry. The conservation law connected with the phase is the conservation of electrical charge. Therefore, it is important to define symmetry.

The state vector and operators are very important concepts in quantum mechanics. A physical state of an atomic system is represented by a state vector $|A\rangle$ which can be made from a linear combination of a complete orthonormal set of states $|i\rangle$ with $i = 1, 2, \dots, n$ as follows:

$$|A\rangle = \sum_i c_i |i\rangle \quad (10)$$

where $c_i = \langle i | A \rangle$ are complex numbers, the amplitudes for the state A to be in the base state i , or the i -components of the state vector A as a dot product of the state vector A and base vector i . The base states i span a vector space of dimension n called Hilbert space.

Performing an operation on a physical state will produce a new state in the vector space. An operator on a vector space is a mapping of the space into itself. It means that an operator Q acting on a state vector A produces a new state vector A' in the space, that is

$$|A'\rangle = Q|A\rangle \quad (11)$$

If it satisfies

$$Q(|A\rangle + |B\rangle) = Q|A\rangle + Q|B\rangle \quad (12)$$

and

$$Q(C|A\rangle) = C Q|A\rangle \quad (13)$$

with C a complex number, then Q is linear. If the transpose conjugate of Q equals to its inverse, that is, $Q^+ = Q^{-1}$, then Q is unitary. When $P^+ = P$, P is hermitian. The unitary operator Q and hermitian operator P are related by

$$Q = e^{i\alpha P} \quad (14)$$

with α a real number.

Theorem 1: The unitary operator of displacement in time from t to t' , $U(t', t)$ is related to the hermitian operator H which is the energy operator of the atomic system as follows:

$$U(t', t) = e^{-\frac{i}{\hbar} H(t'-t)} \quad (15)$$

Proof: Consider the state vector of an atomic system at t' as the result of the operation of the time displacement operator on the state vector at t , that is

$$|A(t')\rangle = U(t', t)|A(t)\rangle \quad (16)$$

For the case of infinitesimal time displacement, $t'-t = \Delta t \rightarrow 0$, $U(t', t) \rightarrow 1$, for small Δt , $U(t+\Delta t, t)$ should differ from unity by an amount proportional to Δt , so that it can be written as

$$U(t+\Delta t, t) = 1 + K\Delta t \quad (17)$$

By substituting into (16), we obtain

$$|A(t')\rangle = (1 + K\Delta t)|A(t)\rangle = |A(t)\rangle + K\Delta t|A(t)\rangle,$$

Or

$$\lim_{\Delta t \rightarrow 0} \frac{|A(t+\Delta t)\rangle - |A(t)\rangle}{\Delta t} = K|A(t)\rangle$$

Consequently,

$$\frac{d}{dt}|A(t)\rangle = K|A(t)\rangle \quad (18)$$

which has a solution

$$|A(t)\rangle = C e^{Kt} \quad (19)$$

Since if a particle at t and at state A has a definite energy E_0 , the amplitude to find it at time t is

$$C e^{-\frac{i}{\hbar} E_0 t} \quad (\text{for a particle at rest})$$

Or

$$C e^{-\frac{i}{\hbar} (Et - px)} \quad (\text{for a particle with momentum } p) \quad (20)$$

By comparison of (19) and (20), therefore, K is just the energy operator, that is

$$K = -\frac{i}{\hbar} H$$

By substituting K in (18), it gives the following important equation of motion known as the Schrödinger equation.

$$i\hbar \frac{d}{dt}|A(t)\rangle = H|A(t)\rangle \quad (21)$$

Or

$$i\hbar \frac{d}{dt} U |A(t_0)\rangle = HU |A(t_0)\rangle$$

Therefore, we get

$$i\hbar \frac{d}{dt} U = HU \quad (22)$$

Which has the following solution

$$U = Ce^{-\frac{i}{\hbar} Ht}$$

When $t=t_0$,

$$U = 1 = Ce^{-\frac{i}{\hbar} Ht_0}, \therefore C = e^{\frac{i}{\hbar} Ht_0} \quad \text{therefore, } U(t, t_0) = e^{-\frac{i}{\hbar} H(t-t_0)}$$

$$\text{or } U(t', t) = e^{-\frac{i}{\hbar} H(t'-t)} \quad (23)$$

QED

Theorem 2: The condition for a physical system to be symmetric under the operation Q is that Q commutes with the Hamiltonian H, that is $QH=HQ$

Proof: Let the Hamiltonian (total energy of the system in the form of operators) and the states of a quantum mechanical system at $t=0$, and t be H, ψ_1 , and ψ_2 .

Then,

$$\psi_2 = U(t, 0)\psi_1 = e^{-\frac{i}{\hbar} Ht}\psi_1 \quad (24)$$

Let Q be a symmetrical operator, then we have

$$Q\psi_1 = \psi_1', \quad Q\psi_2 = \psi_2' \quad (25)$$

and

$$\psi_2' = U\psi_1' \quad (26)$$

$$\text{Substitute (24) and (25) in (26): } Q\psi_2 = UQ\psi_1 \text{ or } QU\psi_1 = UQ\psi_1. \quad (27)$$

$$\text{Since } (QU-UQ)\psi_1=0, \text{ therefore, we have } QU-UQ=0 \text{ or } [Q, U]=0 \quad (28)$$

$[Q, U]=QU-UQ$ is called a commutator. If the commutator is zero, then Q and U

$$U = e^{-\frac{i}{\hbar} Ht} = 1 - \frac{i}{\hbar} H\delta$$

commute. For infinitesimal time $t=\delta$, using Taylor expansion in t, we have

$$QU - UQ = Q(1 - \frac{i}{\hbar} H\delta) - (1 - \frac{i}{\hbar} H\delta)Q = -QH + HQ = 0$$

so that

$$\text{or } HQ - QH = 0, \text{ or } [H, Q] = 0 \quad (29)$$

QED

Theorem 3: A symmetrical operator Q is a constant.

Proof: The time dependence of an operator Q in Heisenberg's picture is given by

When Q does not explicitly depend on time, $\partial Q / \partial t = 0$. Because of symmetry, by Theorem 2, $[H, Q] = 0$

$$\frac{d}{dt} Q = \frac{\partial}{\partial t} Q + \frac{i}{\hbar} [H, Q] \quad (30)$$

$$\therefore \frac{dQ}{dt} = 0, \quad Q = C$$

QED

For $Q=H$, $[H, H]=0$, therefore, H is a constant and energy is conserved.

Similarly, it can be shown that linear momentum and angular momentum are conserved if translation and rotation operations are symmetric respectively.

Theorem 4: If a system is symmetric with respect to an inversion operation, then the parity of the system is conserved.

Proof: Let P be the inversion operator and let the state of the system be $\psi(x, y, z)$, then we have

$$P|\psi(x, y, z)\rangle = |\psi(-x, -y, -z)\rangle = e^{i\alpha}|\psi(x, y, z)\rangle$$

because the inversion operation does not lose any particle of the system so that the normalization of $\psi(x, y, z)$ and $\psi(-x, -y, -z)$ must be the same; they can only differ by a factor $e^{i\alpha}$ with α to be real.

$$|\psi(x, y, z)\rangle = P|\psi(-x, -y, -z)\rangle = e^{i\alpha}P|\psi(x, y, z)\rangle = e^{i\alpha}e^{i\alpha}|\psi(x, y, z)\rangle$$

One more inversion will be right back to the original state, that is

Therefore,

$$(e^{i\alpha})^2 = 1, \text{ or } e^{i\alpha} = \pm 1$$

Which means that

$$P|\psi(x, y, z)\rangle = |\psi(x, y, z)\rangle \text{ or } P|\psi(x, y, z)\rangle = -|\psi(x, y, z)\rangle$$

In classical mechanics, if a state is symmetric under inversion, the operation gives back the same state. In quantum mechanics, it gives back the same state (even parity) or negative of the same state (odd parity). When P and H are commute, then P is a constant, and parity is conserved.

QED

It is true for electromagnetic, gravitational, and strong interactions but not for weak interaction. An experiment proposed by T.D. Lee and C.N. Yang and performed by C.S. Wu and her collaborators in 1956 showed that electrons were emitted preferentially in the direction opposite to the polarized Co-60 nuclei. Therefore, inversion symmetry is violated in beta decay or parity is not conserved in weak interaction. It means that the spin of the emitted electron is anti-parallel to its momentum; such an electron is said to be left-handed.

In addition to electrons there must be positrons that are necessarily related to electrons. The properties of these two particles are: the energies are equal; the masses are equal; the charges are reversed; they can annihilate each other to give photons. It was supposed that matter and antimatter were symmetric, so that the charge conjugation or matter antimatter conjugation operator (C-parity) is conserved. It is true for electromagnetic and strong interactions but not for weak interaction. It was thought that the product CP would be conserved by the weak interaction. If CP is conserved, the long live neutral K_L^0 meson can decay into three pions. But J.W. Cronin and V.L. Fitch found that 0.3% of the neutral K_L^0 meson decayed into two pions in 1964. Therefore, CP is also violated in weak interaction. It becomes very interesting to test the time reversal T invariance in weak interaction, because CPT was supposed to be conserved in general. Time reversal invariance means that physical processes are equally capable of proceeding forward and backward. Recently, it was reported in the Central European de la Recherche Nucleaire (CERN), that the neutral K and the anti-neutral K-mesons did not decay in the same rate into each other. The decay rate of anti-neutral K to neutral K was discovered to be higher than the reverse process. Anti-matter has a shorter lifetime. Therefore, T is also violated in weak interaction. It also explains why we have more matter than anti-matter in our world. Consequently, the study of symmetries and conservation laws of physics is extremely important for us to understand the material world that we are living in.

Consequently, the study of symmetries and conservation laws of physics is extremely important for us to understand the material world that we are living in.

The Study of Axiomatic Properties of the Natural Logarithm Function

Weikai Lang

Non-Euclidean Geometries were a result of changing Euclid's axiom of "The sum of the measures of the interior angles of a triangle is always 180° ." Specifically, they were the changing of the part which stated "the sum of the measures... is always 180° " into "the sum of the measures... can be greater than 180° " and "the sum of the measures... can be less than 180° ".

In this paper, our primary objective is to examine what happens to the natural logarithm function when one of its defining properties is altered. Who knows? It might lead into an "Artificial Logarithm" function.

It is a common result in mathematics that for a function $f(x)$ such that f is continuous and one-to-one in the domain of positive real numbers, with the property that $f(xy) = f(x) + f(y)$, it implies that the function f is a logarithmic function. The meat of my paper will be to find out what happens to the function after I change the property $f(xy) = f(x) + f(y)$ into $f(xy) = f(10x) + f(10y)$.

First, the properties of $f(x)$: (for all $a, b > 0, n \in \mathbb{R}$)

1.

$$f(ab) = f(a) + f(b)$$

2.

$$f(1) = 0$$

3.

$$f\left(\frac{a}{b}\right) = f(a) - f(b)$$

4.

$$f(a^n) = nf(a)$$

Now we begin to study the axiomatic properties of $f(x)$.

| note: Before we start showing the properties of the new function, let's denote it by $\mathcal{G}(x)$.

Analogy-Property 1 of $f(x)$ of $\mathcal{G}(x)$ is given in the definition of $\mathcal{G}(x)$

Showing the Analogy-Property 2 of $\mathcal{G}(x)$ is quite simple, because when we take a and $b = 10$

$$\mathcal{G}(ab) = \mathcal{G}(10a) + \mathcal{G}(10b) \Rightarrow \mathcal{G}(100) = \mathcal{G}(100) + \mathcal{G}(100)$$

$$\mathcal{G}(100) = 2\mathcal{G}(100)$$

and once again, because $\mathcal{G}(x)$ is a one-to-one function,

$$\mathcal{G}(100) = 0$$

(Analogy-Property 2)

Showing the Analogy-Property 3 of $f(x)$ of $\mathcal{G}(x)$ starts out very similar to, but has a little twist at the end, because we start with

$$\mathcal{G}\left(\frac{a}{b}\right) = \mathcal{G}(a \times b^{-1}) = \mathcal{G}(10a) + \mathcal{G}(10(b^{-1}))$$

and then go on to

$$\begin{aligned}\mathcal{G}(100) &= \mathcal{G}(100(1)) = \mathcal{G}(10b \times 10b^{-1}) = \mathcal{G}(100b) + \mathcal{G}(100b^{-1}) = 0 \\ \mathcal{G}(100b^{-1}) &= \mathcal{G}\left(100\left(\frac{1}{b}\right)\right) = -\mathcal{G}(100b)\end{aligned}$$

which is acceptable because that results in

$$\mathcal{G}\left(\frac{a}{b}\right) = \mathcal{G}(10a) - \mathcal{G}(1,000b)$$

However, $f(x^{-1}) = -f(x)$ is required in the proof of $f(x)$, so it hardly seems fair if we don't show what $\mathcal{G}(x^{-1})$ equals in our Analogy-Property 3. So, in the interests of following the techniques of showing the properties of $f(x)$ as close as possible, we start with which is Analogy-Property 3, but we also notice that

$$\begin{aligned}\mathcal{G}\left(\frac{a}{b}\right) &= \mathcal{G}(10a) - \mathcal{G}(1,000b) \\ -\mathcal{G}\left(\frac{a}{b}\right) &= \mathcal{G}(1,000b) - \mathcal{G}(10a)\end{aligned}$$

which is important because

$$\begin{aligned}\mathcal{G}\left(\frac{100b}{\left(\frac{a}{100}\right)}\right) &= \mathcal{G}(1,000b) - \mathcal{G}(10a) \\ \mathcal{G}\left(\frac{100b}{\left(\frac{a}{100}\right)}\right) &= \mathcal{G}\left(10,000\frac{b}{a}\right) = -\mathcal{G}\left(\frac{a}{b}\right) \\ -\mathcal{G}\left(\frac{a}{b}\right) &= \mathcal{G}\left(\frac{100b}{\left(\frac{a}{100}\right)}\right)\end{aligned}$$

which is elegant because that gives us the relation

$$\mathcal{G}(x^{-1}) = -\mathcal{G}(10,000x)$$

(Analogy-Property 3.1)

Now we set out to find Analogy-Property 4 for $\mathcal{G}(x)$. Before we actually proceed, it is important to note that we want the findings to be in the form

$$\mathcal{G}(x^n) = \dots$$

not only for the sake of attempting to parallel the properties of $f(x)$, but also because of its ease in implementation.

¹ for $n \in \mathbb{N}$,

$$\mathcal{G}(x^n) = \underbrace{\mathcal{G}(x \times x \times \dots \times x)}_{n \text{ times}} = \underbrace{\mathcal{G}(10x) + \mathcal{G}(10x) + \dots + \mathcal{G}(10x)}_{n \text{ times}} = n\mathcal{G}(10x)$$

$$\mathcal{G}(x^n) = n\mathcal{G}(10x)$$

(Analogy-Property 4.1)

II for $n \in \mathbb{Z}$, $n < 0$
 setting $n = -m$

$$\mathcal{G}(x^{-m}) = \mathcal{G}\left(\underbrace{\frac{1}{x} \times \frac{1}{x} \times \dots \times \frac{1}{x}}_{m \text{ times}}\right) = \mathcal{G}\left(\frac{10}{x}\right) + \mathcal{G}\left(\frac{10}{x}\right) + \dots + \mathcal{G}\left(\frac{10}{x}\right) = m\mathcal{G}\left(\frac{10}{x}\right)$$

using Analogy-Property 3.1, we have

$$m\mathcal{G}\left(\frac{10}{x}\right) = -m\mathcal{G}(1,000x)$$

which leads to

$$\mathcal{G}(x^{-m}) = -m\mathcal{G}(1,000x)$$

(Analogy-Property 4.2)

III $n \in \mathbb{Q}$, $n \neq 0$

A. (remember we have the property in the form $\mathcal{G}(x^n) = \dots$; so we start with...)

$$\mathcal{G}(x^{\frac{1}{n}}) = \dots$$

$$\mathcal{G}\left(\frac{x}{10^n}\right) = \mathcal{G}\left(\underbrace{\frac{x^n}{10} \times \frac{x^n}{10} \times \dots \times \frac{x^n}{10}}_{n \text{ times}}\right) = \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) + \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) + \dots + \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) = n\mathcal{G}\left(\frac{x^n}{10}\right)$$

$$\mathcal{G}(x^{\frac{1}{n}}) = \frac{1}{n}\mathcal{G}\left(\frac{x}{10^n}\right)$$

(Analogy-Property 4.3a)

B.

$$\mathcal{G}(x^{\frac{m}{n}}) = \dots$$

$$\mathcal{G}\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = \mathcal{G}\left(\underbrace{\frac{x^n}{10} \times \frac{x^n}{10} \times \dots \times \frac{x^n}{10}}_{m \text{ times}}\right) = \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) + \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) + \dots + \mathcal{G}\left(10\left(\frac{x^n}{10}\right)\right) = m\mathcal{G}\left(\frac{x^n}{10}\right)$$

$$\mathcal{G}\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = m\mathcal{G}\left(x^{\frac{1}{n}}\right) = m\mathcal{G}\left(\frac{1}{n}\left(\frac{x}{10^n}\right)\right)$$

$$\mathcal{G}\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = \frac{m}{n}\mathcal{G}\left(\frac{x}{10^n}\right)$$

(equation 1)

At first glance, this might look like Analogy-Property 4.3b, but it's not in the form which we want. In order to arrive at that form, we need to make

$$\vartheta\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = \vartheta\left(\frac{x^{\frac{m}{n}}}{10} \times \frac{1}{10^{m-1}}\right) = \vartheta\left(10\left(\frac{x^{\frac{m}{n}}}{10}\right)\right) + \vartheta\left(\frac{10}{10^{m-1}}\right) = \vartheta\left(x^{\frac{m}{n}}\right) + \vartheta\left(\frac{1}{10^m}\right)$$

using Analogy-Property 3.1, we have

$$\vartheta\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = \vartheta\left(x^{\frac{m}{n}}\right) + \vartheta\left(\frac{1}{10^m}\right) = \vartheta\left(x^{\frac{m}{n}}\right) - \vartheta(10,000(10^m)) = \vartheta\left(x^{\frac{m}{n}}\right) - \vartheta(10^{m+4})$$

(equation 2)

Now we can start our final approach to Analogy-Property 4.3b, using equations 1 & 2 we get

$$\vartheta\left(x^{\frac{m}{n}}\right) - \vartheta(10^{m+4}) = \vartheta\left(\frac{x^{\frac{m}{n}}}{10^m}\right) = \frac{m}{n} \vartheta\left(\frac{x}{10^n}\right)$$

which leads us to

$$\vartheta\left(x^{\frac{m}{n}}\right) = \frac{m}{n} \vartheta\left(\frac{x}{10^n}\right) + \vartheta(10^{m+4})$$

(Analogy-Property 4.3b)

C. now for $n < 0$

We can't approach this the same way as $f(x)$. Let's start with

$$\vartheta\left(\frac{x^{-1}}{10^n}\right) = \vartheta\left(\underbrace{\frac{x^{-\frac{1}{n}}}{10} \times \frac{x^{-\frac{1}{n}}}{10} \times \dots \times \frac{x^{-\frac{1}{n}}}{10}}_{n \text{ times}}\right) = \underbrace{\vartheta\left(10\left(\frac{x^{-\frac{1}{n}}}{10}\right)\right) + \vartheta\left(10\left(\frac{x^{-\frac{1}{n}}}{10}\right)\right) + \dots + \vartheta\left(10\left(\frac{x^{-\frac{1}{n}}}{10}\right)\right)}_{n \text{ times}} = n \vartheta\left(x^{-\frac{1}{n}}\right)$$

$$\vartheta\left(x^{-\frac{1}{n}}\right) = \frac{1}{n} \vartheta\left(\frac{x^{-1}}{10^n}\right)$$

Now for

$$\vartheta\left(x^{-\frac{m}{n}}\right) = ?$$

(from Analogy-Property 3.1)

$$\begin{aligned} \vartheta\left(x^{-\frac{m}{n}}\right) &= \vartheta\left(\left(x^{\frac{m}{n}}\right)^{-1}\right) = -\vartheta(10,000x^{\frac{m}{n}}) \\ -\vartheta(10,000x^{\frac{m}{n}}) &= -\vartheta\left(100,000\left(\frac{x^{\frac{m}{n}}}{10}\right)\right) = -\vartheta\left(x^{\frac{m}{n}}\right) \\ -\vartheta\left(x^{\frac{m}{n}}\right) &= -\left[\frac{m}{n} \vartheta\left(\frac{x}{10^n}\right) + \vartheta(10^{m+4})\right] \\ \vartheta\left(x^{-\frac{m}{n}}\right) &= -\frac{m}{n} \vartheta\left(\frac{x}{10^n}\right) - \vartheta(10^{m+4}) \end{aligned}$$

(Analogy-Property 4.3c)

IV $n \in \mathbb{I}$

Let $x \in \mathbb{I}$, and $x_n \in \mathbb{Q}$, and x_n can be expressed as

$$x_n = \frac{m}{n}$$

Therefore, Analogy-Property 4.3b can be expressed as

$$\vartheta(a^{x_n}) = x_n \vartheta\left(\frac{a}{\frac{m}{n}}\right) + \vartheta(10^{nx_n+4})$$

Now, because $\vartheta(x)$ is continuous,

$$\lim_{x_n \rightarrow x} \vartheta(a^{x_n}) = \vartheta(\lim_{x_n \rightarrow x} a^{x_n}) = \vartheta(a^x)$$

Also,

$$\lim_{x_n \rightarrow x} \vartheta(a^{x_n}) = \lim_{x_n \rightarrow x} [x_n \vartheta\left(\frac{a}{\frac{m}{n}}\right) + \vartheta(10^{nx_n+4})]$$

$$\lim_{x_n \rightarrow x} \vartheta(a^{x_n}) = \lim_{x_n \rightarrow x} [x_n \vartheta\left(\frac{a}{\frac{m}{n}}\right) + \vartheta(10^{nx_n+4})] = \left(\lim_{x_n \rightarrow x} x_n\right) \left[\vartheta\left(\lim_{x_n \rightarrow x} \frac{a}{\frac{m}{n}}\right)\right] + \vartheta\left(\lim_{x_n \rightarrow x} 10^{nx_n+4}\right)$$

$$\lim_{x_n \rightarrow x} \vartheta(a^{x_n}) = x \vartheta\left(\frac{a}{\frac{m}{x}}\right) + \vartheta(10^{nx+4})$$

$$\therefore \vartheta(a^x) = x \vartheta\left(\frac{a}{\frac{m}{x}}\right) + \vartheta(10^{nx+4})$$

(Analogy-Property 4.4)

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